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SECTION 2

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GROUP 17

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TABLE OF CONTENTS

1.0 Project Overview	3
1.1 Description on the Solution.....	3
2.0 Technology Use	4
3.0 Data Pipeline Architecture	4
4.0 MS Power BI dashboard	5
5.0 Reflection	6

1.0 Project Overview

In this Microsoft Azure tutorial, we build a comprehensive data pipeline on Azure. The core objective of this tutorial is to address a critical business which is to determine a gap in the knowledge about customer demographics, specifically the gender distribution of the customer population and analyzing how it influences product purchases. We extract the customer and sales data from an on-premises SQL database, transform it in the cloud, and generate actionable insights through a Power BI dashboard. The dashboard will highlight key performance indicators (KPIs) related to gender distribution and product category sales, allowing stakeholders to filter and analyze data by date, product category, and gender. The pipeline makes sure that the business can access accurate and up-to-date data at any time without requiring manual intervention.

1.1 Description on the Solution

To process the data efficiently and maintain a clean organizational structure, the solution implements a Medallion Architecture with Azure Data Lake Storage Gen2. This framework logically segregates data into three layer

Bronze Layer: Azure Data Factory (ADF) extracts the data directly from the on-premise SQL database and dumps it into the Bronze container in Parquet format. The data is completely raw and unmodified.

Silver Layer: In this Azure Databricks perform minor data cleansing and transformations. For example, messy timestamp formats containing unnecessary time and timezone data from the Bronze layer are parsed and standardized into a clean yyyy-MM-dd date format. This cleansed data is then saved in the highly optimized Delta format.

Gold Layer: This is the final presentation layer for analytical querying and reporting. Databricks applies final structural transformations to the Silver data, such as converting all column names from PascalCase to snake_case like changing ModifiedDate to modified_date to make them more query-friendly. Azure Synapse Analytics reads the refined Delta tables and creates logical SQL views to serve the data seamlessly to Power BI.

2.0 Technology Use

This project has utilized a number of Microsoft Azure technologies to create an end-to-end data engineering pipeline of customer and sales data. All of the services have been chosen according to the contribution to the workflow, such as data ingestion, secure connectivity, storage, transformation, analytics, and reporting.

The source system was **Microsoft SQL server** to simulate an on-premise operational database. It hosted AdventureWorksLT2025 database comprising customer, product, and sales related tables. This is a realistic business situation whereby transactional data is initially stored in relational systems and then incorporated into a cloud analytics environment.

Azure Data Factory (ADF) was used as the main orchestration and ingestion service. It linked to the SQL Server source, and downloaded the chosen tables to the storage of Azure. ADF was selected as an efficient low-code tool to automate the data movement and allows to run pipelines on a schedule. As the SQL Server source was on-premises, a Self-hosted Integration Runtime (SHIR) was also needed to enable safe communication between the on-premises and Azure environments.

Sensitive credentials (SQL Server password) were stored in **Azure Key Vault**. This enhanced security since credentials were not hardcoded in the pipeline. It is also a good governance practice, since the authentication information was not attached to the pipeline logic.

The storage layer of the solution was **Azure Data Lake Storage Gen2 (ADLS Gen2)**. Data was classified into Bronze, Silver and Gold layers. Raw ingested data was stored in the Bronze layer, cleaned and standardised data was stored in the Silver layer and analytics-ready data was stored in the Gold layer. This stratified design enhanced maintainability and simplified the management of the pipeline.

Data transformation was done using **Azure Databricks**. Data were cleaned, standardised (e.g. date fields) and renamed columns were converted into more analysis-friendly formats before the processed data were written to the Silver and Gold layers using databricks notebooks. A

structured analytics layer was then provided by using **Azure Synapse Analytics** to structure the final processed data into queryable views.

Lastly, dashboards and visual reports were created using **Microsoft Power BI**. It showed major business data including total sales, number of products and customer trends using interactive visualisations. All in all, the chosen technologies were synergistic and complemented each other to facilitate the entire data pipeline between the raw source data and decision-making insights.

Figure 1 illustrates the overall architecture of the proposed Azure-based data engineering solution. Data is first extracted from the on-premises SQL database using Azure Data Factory, which acts as the ingestion and orchestration service. The extracted data is stored in Azure Data Lake Storage and organised into Bronze, Silver, and Gold layers to separate raw, cleaned, and analytics-ready datasets. Azure Databricks is used to transform and standardise the data between these layers. The processed Gold-layer data is then accessed through Azure Synapse Analytics to support structured querying and reporting. Finally, Microsoft Power BI is used to create dashboards and visual insights for business users. Security and governance are supported through Azure Active Directory for identity management and Azure Key Vault for secure credential storage.

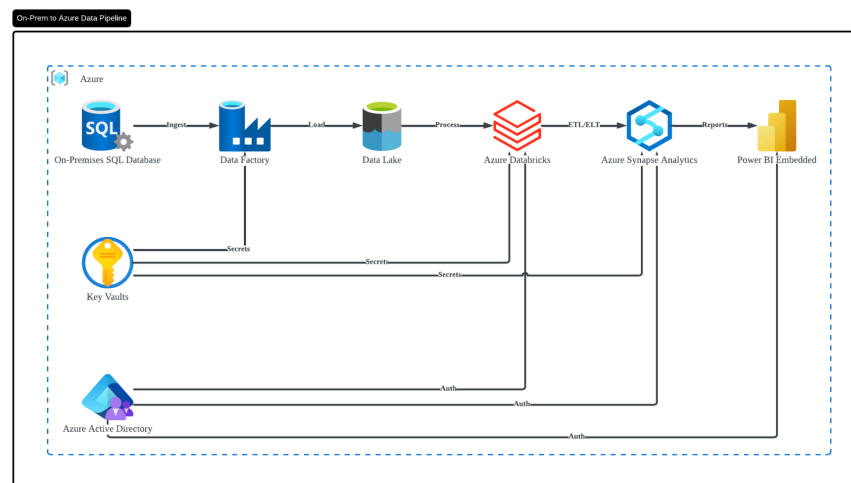
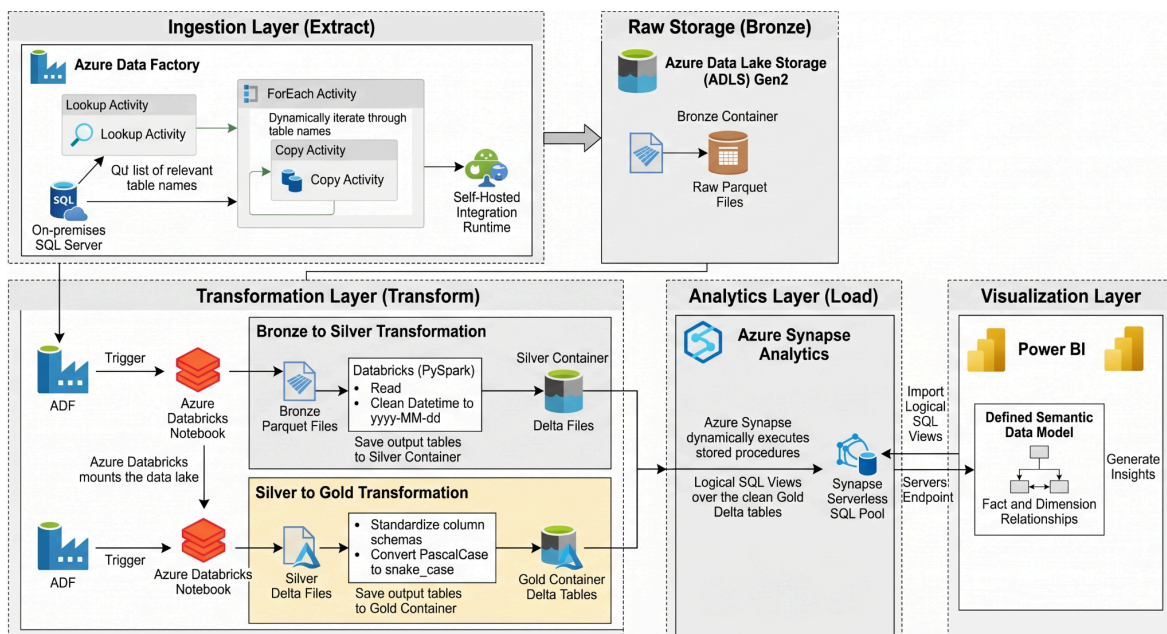


Figure 1 : Azure Data Pipeline Architecture

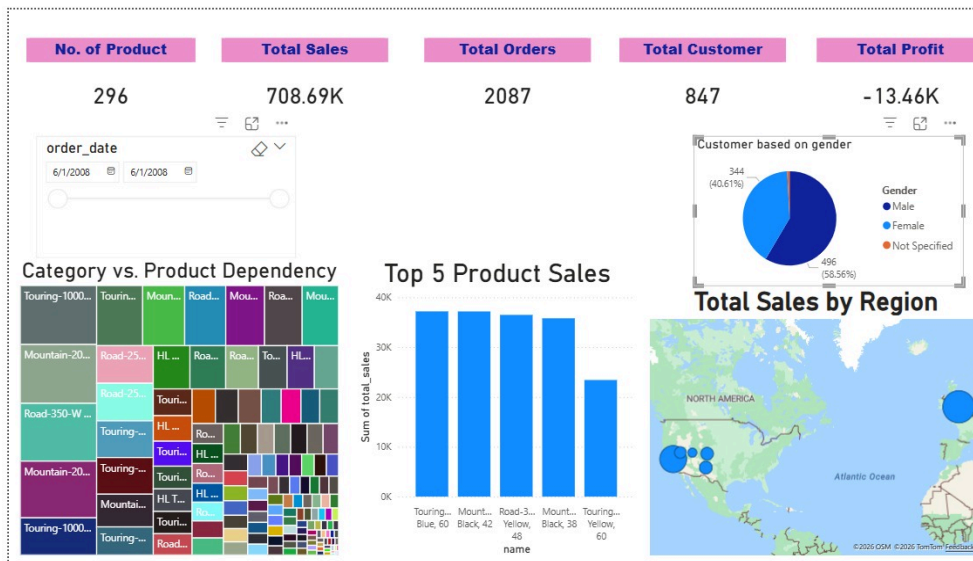
3.0 Data Pipeline Architecture

The architecture follows the standard Extraction, Transform, Loading (ETL) workflow.

Ingestion Layer (Extract)	Azure Data Factory will trigger a pipeline containing a Query activity to query the name list of all related tables from the SQL server on the internal server. Subsequently, the For Each activity will dynamically traverse these tables and replicate the data at runtime through self-managed integration.	
Raw Storage (Bronze)	The data obtained will be stored directly in the Bronze container of ADLS in the original parquet file format.	
Transformation Layer (Transform)	Bronze to Silver	The PySpark program will convert the chaotic date-time values into a standardised date format (yyyy-MM-dd), and then save these tables in the Silver container in the form of Delta files.
	Silver to Gold	The second Databricks job will standardise the column format, convert the columns in PascalCase format to snake_case format, and save the final Delta table in the Gold container.
Analytics Layer (Load)	Azure Synapse will dynamically execute the storage process to create a logical SQL view on a clean Gold data table.	
Visualisation Layer	Power BI will import these views from Synapse serverless endpoints and use predefined semantic data models like handling the relationship between fact data and dimensional data to generate insights.	



4.0 MS Power BI dashboard



KPI Cards	The top of the dashboard allows users to immediately see key business indicators. It shows data such as the number of products (296) and total sales (7,086,900) according to the requirements of stakeholders. However, it further provides more in-depth background information, including the total number of orders (2,087), the total number of customers (847) and the total profit (-134,600).
Demographic Insights	In order to achieve the main business goal of understanding the customer base, a pie chart called Customer Group By Gender shows the distribution of the population structure. The chart shows that men dominate the customer group (496 customers, accounting for 58.56%), followed by women (344 customers, accounting for 40.61%). The rest is not specified.
Product Performance	The Five Best-selling Products bar chart can quickly highlight the best-performing products, whilst the Category and Product Dependency tree chart can show the product hierarchy in detail like tourism category, mountain category, road category so that stakeholders can intuitively understand which categories dominate inventory and sales.
Geographical Analysis	The Total Sales by Region chart visualisation graphically presents the sales distribution in North America and Europe through the size of bubbles, adding valuable geographical dimensional information to the sales data.
Data Storytelling	In order to meet the requirements of a user-friendly interface, we have added an interactive order date filter. This allows stakeholders to dynamically filter the entire dashboard according to a specific time range. For example, only data from June 2008 is displayed and update all charts immediately to respond to queries based on a specific date.

5.0 Reflection

CHOH JING YI, A23CS0296

I learned how to build an end to end pipeline on Microsoft Azure. I followed the youtube tutorial to extract the raw data from the on premise database to clean and finally create a visual dashboard with power BI. I understand the Medallion Architecture which transforms step-by-step from Bronze (raw) to Silver (clean) to Gold (business-ready). I also learn practical skills in Azure Data Factory (ADF), PySpark in Databricks, and Azure Synapse Analytics. I face some errors when configuring the connectivity and compute environments. I struggled with the "Login failed" error when connecting to the Azure SQL Database. I learned what is Fully Qualified Domain Name (FQDN) and realized that UTM WiFi often uses shifting IP addresses. So I need to configure server-level firewall rules and allow Azure services to bypass the gateway. Overall, this project taught me how to build an end to end data pipeline using Microsoft Azure.

LAU YAN KAI, A23CS0098

In the process of building this end-to-end pipeline, some technical challenges were encountered by us. First of all, connecting the cloud-based Azure data factory with the local SQL server requires downloading and configuring specific self-hosted integrated runtime to act as a security bridge. In addition, during the configuration of Databricks, we encountered restrictions related to the transmission of credentials in the multi-node cluster of the trial account. We solved this problem by switching to a single-node cluster. Overall, the project has greatly improved our practical understanding of modern cloud data engineering. We learned how to transform theoretical concepts such as bronze-silver-gold medallion layout into practical application. We also have an in-depth understanding of automated security and learn how to use managed identities and Azure Key Vault instead of hard-coded passwords.

TAN ZHI MING, A23CS0189

This project provided me with a real-life experience of constructing an actual end-to-end data engineering pipeline in Azure. Not only did I get to understand how services like Azure Data Factory, Data Lake, Databricks, Synapse, and Power BI can be used in conjunction with each other, but also the significance of troubleshooting, security, and configuration in cloud-based systems. Connection and authentication were one of the key issues to overcome between the on-premises SQL Server and Azure, and finding solutions to these issues helped me to gain knowledge about technical aspects and problem solving. All in all, this project allowed me to bridge the gap between theory and practice and provided me with a better idea of how data engineering can assist business insight and decision-making.